

Error Analysis Report

on Heart Failure Prediction Dataset

1. Three Concrete Model Mistakes

Input Example	True Label	Model Prediction	Failure Analysis
Age: 48, ChestPain: 'ASY', RestingBP: 138, Cholesterol: 214 (No5)	Disease (1)	Healthy (0)	False Negative (FN): Clinical markers appeared near-normal in this particular dataset, masking the real issues the model cannot spot yet
Age: 67, ChestPain: 'ASY', RestingBP: 160, Cholesterol: 286 (No 886)	Disease (1)	Healthy (0)	False Negative (FN): High age is often linked with high blood pressure; the model struggle from a skewed training set
Age: 43, ChestPain: 'ASY', RestingBP: 132, Cholesterol: 247 (No 889)	Disease (1)	Healthy (0)	Low Sensitivity: The very low <i>Oldpeak</i> value (0.1) failed to surpass the default probability threshold of 0.5, leading to false prediction.

2. Why it may have failed

- **Limited Feature Scope:** Clinical markers often look "normal" even when a patient is ill. The model struggles here because it lacks info on hidden risk factors—like family history or lifestyle—that fall outside the scope of our current dataset.
- **Conservative Prediction Bias:** Even with our preprocessing, the model remains cautious, with a tendency to default to the "Healthy" class whenever the input features aren't very high, probably because of some leftover class imbalance in the training data.
- **Data Quality & Overlap Issues:** The dataset suffers from significant noise, particularly with 0 values for Cholesterol and RestingBP - which are clinically impossible. Furthermore, these variables show big overlap between the 'Healthy' and 'Disease' groups, especially in older patients, which makes it very challenging for our model to make a clear decision boundary, since it is confused by the mixed patterns and outliers in the data."

3. What should be tried next (to be fixed)

- **Threshold Tuning:** Instead of the default 0.5 threshold, we will optimize the classification threshold using a Precision-Recall curve to prioritize **Recall**, thereby minimizing the rate of missed diagnoses (False Negatives).
- **Interaction Features:** Engineer new features by combining *Age* with *MaxHR* or *Oldpeak* to better highlight pathological correlations, making the model more sensitive to borderline cases.
- **Ensemble Techniques:** Transition from Logistic Regression to more robust ensemble methods, such as Random Forest or XGBoost with *class_weight='balanced'*, to effectively handle data noise and non-linear patterns.